

Research paper

Data-driven subtypes of internalizing and functional somatic symptoms: A hybrid mixture modeling approach



Urvi Saini^{a,*}, Rob Wanders^b, Albertine J. Oldehinkel^a, Judith G.M. Rosmalen^{a,c},
Hanna M. van Loo^a

^a University of Groningen, University Medical Center Groningen, Department of Psychiatry, Interdisciplinary Center Psychopathology and Emotion Regulation (ICPE), Groningen, the Netherlands

^b Parnas, Groningen, the Netherlands

^c University of Groningen, University Medical Center Groningen, Department of Internal Medicine, Groningen, the Netherlands

ARTICLE INFO

Keywords:

Depression
Anxiety
Chronic fatigue syndrome
Fibromyalgia
Irritable bowel syndrome
Mixture item response theory
Functional somatic disorders

ABSTRACT

Background: The mechanisms underlying the high comorbidity between internalizing disorders (IDs) and functional disorders (FDs) remain unclear. This study aimed to identify data-driven subgroups of major depressive disorder (MDD), generalized anxiety disorder (GAD), myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS), fibromyalgia (FM), and irritable bowel syndrome (IBS) symptoms in the general population, capturing shared symptom patterns while accounting for variation in symptom severity.

Method: We analyzed cross-sectional data from 72,919 adults in the Dutch Lifelines Cohort Study, randomly divided into training and validation subsets. Twenty-seven ID and FD criteria symptoms were examined using mixed-measurement item response theory models in the training set, with the optimal model validated in the validation set. Class characteristics were assessed through associations with risk factors, ID/FD diagnosis and comorbidity patterns.

Results: Six classes best described the data: Healthy (57.5%), Pain (13.8%), Tension/Pain (9.1%), Anxiety (9.0%), Cognition/Fatigue (6.4%), and Depression (4.3%). All classes showed a combination of ID and FD symptoms, with the Cognition/Fatigue and Depression classes being most mixed, and the Pain and Anxiety classes most domain-specific. Transdiagnostic symptoms were highly endorsed across all classes. ID–FD comorbidity was highest in the Cognition/Fatigue and Depression classes, with the Depression class showing the greatest functional impairment.

Conclusions: The identified classes comprised mixed symptoms from multiple disorders, rather than disorder-specific profiles. ID–FD comorbidity may partly result from non-specific defining symptoms lowering the threshold for fulfilling criteria for multiple disorders. Symptom-level evaluation can help identify more homogenous and clinically relevant subgroups. Longitudinal research is needed to clarify underlying mechanisms.

1. Introduction

Internalizing disorders (IDs) are characterized by emotional and behavioral responses directed inward rather than toward others [1]. The two most common IDs, major depressive disorder (MDD) and generalized anxiety disorder (GAD), frequently co-occur, resulting in greater social and occupational impairment [2,3]. Functional disorders (FDs) involve persistent physical symptoms not attributable to known identifiable pathological abnormalities [4], such as myalgic

encephalomyelitis/chronic fatigue syndrome (ME/CFS), fibromyalgia (FM), and irritable bowel syndrome (IBS). IDs and FDs are highly comorbid [5], and their co-occurrence is associated with reduced quality of life [6]. The mechanisms underlying this comorbidity remain unclear, although genetic, biological, and symptom overlap explanations have been proposed [7–12].

The heterogenous nature of these disorders may contribute to the challenges in uncovering mechanisms underlying ID–FD comorbidity [13–15]. Categorical diagnoses may obscure important variation, as

* Corresponding author at: University of Groningen, University Medical Center Groningen, Interdisciplinary Center Psychopathology and Emotion regulation, Hanzeplein 1, PO Box 30001, 9700, RB, Groningen, the Netherlands.

E-mail address: u.saini@umcg.nl (U. Saini).

<https://doi.org/10.1016/j.genhospsych.2026.08.006>

Received 15 June 2026; Received in revised form 6 August 2026; Accepted 6 August 2026

Available online 11 August 2026

0163-8343/© 2026 The Author(s). Published by Elsevier Inc. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

individuals with the same diagnosis but different symptom patterns and severity may not share the same underlying mechanisms. Joint investigation of ID and FD symptoms to identify more homogenous subgroups might help reveal shared pathways underlying co-occurrence and disorder-specific mechanisms, including genetic or environmental risk factors. Previous studies using latent class models (LCMs) within IDs or FDs have identified cross-diagnostic subgroups with distinct psychological and somatic characteristics, risk factors, and treatment responses [16–20], and symptom profiles that were mixed rather than “pure”. Extending this approach across ID and FD symptoms may help uncover cross-disorder symptom patterns that improve understanding of their comorbidity.

Traditional LCMs impose categorical distinctions that overlook within- and between-class severity variation and the dimensional nature of psychopathology. When the underlying structure is dimensional, increasing the number of classes may approximate severity differences, producing subgroups that do not differ qualitatively and limit interpretability. Symptoms are also often interconnected, potentially violating the conditional independence assumption [21]. A previous factor analysis study showed that ID and FD symptoms naturally cluster along continuous, overlapping dimensions rather than discrete categories [10]. Hybrid mixture models, such as mixed-measurement item response theory (MM-IRT), address these limitations by combining categorical classes with dimensional latent traits, identifying subgroups that differ in both symptom patterns and underlying severity. Several studies show that hybrid mixture models outperform traditional LCMs in identifying more meaningful and clinically informative subgroups, demonstrating improved model fit with fewer classes and avoiding solutions that merely reflect severity gradients [22–24]. Understanding patterns of co-occurring ID and FD symptoms may support more integrated care strategies for patients who do not fit single-diagnosis frameworks. Additionally, empirically derived subtypes may identify more homogenous groups with specific prognosis and treatment response than current classifications.

Existing hybrid model studies have primarily focused on mood and anxiety disorders, with none applying these models to understand ID-FD comorbidity. In this explorative study, we aimed to identify data-driven subgroups based on ID and FD symptom patterns in the general population. As a secondary aim, we characterized these subgroups by examining associations with disability and external variables to describe class differences in clinically relevant characteristics [25–27]. We also examined the distribution of ID and FD cases and their comorbidity patterns across subgroups to assess how established diagnoses relate to empirically derived symptom profiles.

2. Methods

2.1. Participants

Lifelines is a multidisciplinary prospective population-based cohort study examining the health and health-related behaviors of 167,729 persons living in the North of the Netherlands in a unique three-generation design [28]. It employs a broad range of investigative procedures in assessing the biomedical, socio-demographic, behavioral, physical and psychological factors that contribute to the health and disease of the general population, with a special focus on multimorbidity and complex genetics. For a detailed description of Lifelines' recruitment procedure, we refer to previous studies [28–30]. The Lifelines cohort study has been approved by the Medical Ethical Committee of the University Medical Center Groningen.

All participants provided written informed consent. We used data from 72,919 participants, ≥ 18 years old, collected during the second assessment wave of Lifelines, between 2014 and 2019. All included participants had available symptom data for both IDs and FDs.

2.2. Measures

2.2.1. Internalizing symptoms

The Mini International Neuropsychiatric Interview (MINI) was used to assess MDD and GAD symptoms as defined by the fourth edition of the Diagnostic and Statistical Manual of Mental Disorders (DSM-IV) (Table Supplement 1(S1)) [31]. MDD symptoms, assessed during the past two weeks, included depressed mood, anhedonia, appetite change, weight gain, weight loss, trouble sleeping, psychomotor retardation, psychomotor agitation, fatigue, worthlessness/guilt, concentration difficulty, and suicidal ideation. GAD symptoms, assessed during the past six months, included excessive worry/anxiety, restlessness, muscle tension, fatigue, concentration difficulty, irritability, and trouble sleeping. All items were self-reported as present (1) or absent (0).

2.2.2. Functional somatic symptoms

ME/CFS symptoms were assessed using the 1994 Centers for Disease Control and Prevention (CDC) criteria (Table S2) [32]. Symptoms, referring to the past six months, included fatigue, sore throat, tender lymph nodes, muscle pain, joint pain, headaches, unrefreshing sleep, post-exertional malaise (PEM), and impaired memory or concentration. Memory and concentration were combined into a single symptom ('concentration difficulty'), as only one is required by the CDC criteria.

FM symptoms were measured using the 2010 American College of Rheumatology (ACR) criteria (Tables S3–S4) [33]. Symptoms included ≥ 3 months of musculoskeletal pain and symptoms measured using the Widespread Pain Index (WPI) [33] and the Checklist Individual Strength (CIS) [34]. The WPI measured 19 body areas with pain during the past week and the CIS measured fatigue, difficulty thinking, concentration difficulty, and unrefreshing sleep in the past two weeks.

IBS symptoms were examined using the Rome III IBS Diagnostic Questionnaire [35], later adjusted to align with the Rome IV criteria [36]. Symptoms included recurrent abdominal pain/discomfort (≥ 1 day/week for ≥ 6 months) and three conditional symptoms: related to defecation, change in stool frequency, and change in stool form, only assessed among participants reporting abdominal pain/discomfort. For the analysis, symptoms were combined into a single item, scored 1 if participants reported abdominal pain/discomfort and at least two conditional symptoms, and 0 otherwise.

FD symptoms were assessed using self-reported questionnaires and dichotomized to reduce estimation complexity and minimize bias that may arise from unevenly distributed thresholds in low prevalence symptoms [37].

2.3. External variables

We examined the distribution of variables across the identified subgroups, selected based on prior evidence demonstrating links with both IDs and FDs [10]: (a) Sociodemographics: sex, age [5,38–41], and work status (currently employed, unemployed and searching for work, unemployed due to disability) [42,43]; (b) Lifestyle: body mass index (BMI) [39,40,44] and heavy drinking [27,45,46]; (c) Environmental adversity: chronic stress in the past year (e.g., work, home, relationship stress) [10,27,47], acute stress in the past year (e.g., severe injury, death of a loved one, divorce) [27,48,49], and childhood sexual abuse [25,50–52]; (d) Self-rated health [53,54]; and (e) Disability: number of days in the past 30 days during which participants were unable to work, bed bound, limited in work/activities, or less focused due to their physical or mental health [55–58]. The external variables are further detailed in the Supplementary Methods.

In addition, we examined the distribution of ID and FD diagnoses and their comorbidity across subgroups to explore how different symptom profiles may correspond to specific disorders or distinct comorbidity patterns.

2.4. Statistical methods

2.4.1. Symptom aggregation

Diagnostic criteria for IDs and FDs include overlapping symptoms, which may lead to spurious or difficult to interpret classes due to local dependence and indicator redundancy [59]. These transdiagnostic symptoms and their associated disorders were trouble sleeping (MDD, GAD), unrefreshing sleep (ME/CFS, FM), fatigue (MDD, GAD, ME/CFS, FM), and concentration difficulty (MDD, GAD, ME/CFS, FM). Following previous studies [10,11], these symptoms were combined to improve model stabilization and class separation. If a symptom was reported as present in at least one measure, the combined symptom was coded as 1 (present). The 19 WPI pain location items were summed and dichotomized using the cut-off defined in the FM criteria, as the number, not location, of pain sites is diagnostically relevant [33] (Supplementary Methods).

2.4.2. Missing data

Missing data on ID and FD symptoms (<0.3%) (Table S5) and external variables (<37.5%) were handled with the full information maximum likelihood method, using all available item response information for each case to provide stable and unbiased model parameter estimates [60].

2.4.3. Models

As a preliminary step, we performed exploratory factor analysis (EFA) and latent class analysis (LCA) to explore the general structure of the 27 ID and FD symptoms and provide a baseline for comparison with the hybrid models. EFA was used to check the tenability of a general severity dimension across symptoms. LCA identifies mutually exclusive subgroups based on response patterns, assuming item responses are conditionally independent given class membership [61].

Subsequently, we estimated the MM-IRT models, combining features of LCA (i.e., identifying unobserved subpopulations with distinct response patterns) and unidimensional IRT (i.e., modeling a continuous latent factor to estimate individual differences in severity). Each item was parametrized by discrimination (capturing how strongly the item relates to the latent trait, i.e., severity), and difficulty (severity threshold for symptom endorsement) parameters [62,63]. MM-IRT captures both class-specific symptom patterns and individual severity differences and relaxes the conditional independence assumption of LCA by allowing within-class symptom associations [64].

2.4.4. Estimation procedure

Data were randomly split into a training set ($n = 36,459$) for fitting the EFA, LCA, and MM-IRT models and a validation set ($n = 36,460$) for validating the stability and generalizability of the final MM-IRT model. Models were estimated by running 50 random start values, up to 700 iterations per set, 200 Expectation-Maximization iterations, and 80 Newton-Raphson iterations to reduce convergence issues and local maxima. Model selection was based on interpretability and goodness-of-fit statistics [65], including the Bayesian information criterion (BIC) [66], sample size adjusted BIC (sBIC) [67], Akaike information criterion (AIC) [68], and consistent AIC (CAIC) [69]. Model classification quality was also assessed. The final MM-IRT model was estimated in the validation set without external variables, after which external variables were included as distal outcomes to estimate their class-specific distributions. As standard approaches may underestimate associations due to classification error, a bias-adjusted three-step method was used [70–72]. EFA was performed in Mplus version 8.2 [73] and LCA and MM-IRT models were fitted in LatentGOLD 6.0 [74].

3. Results

3.1. Sample description

Of the 72,919 participants included, 59.2% were female and the mean age was 49.1 years ($SD = 12.4$) (Table 1). No significant differences were observed between the training and validation set (Table S6). Overall, 15.4% of participants met diagnostic criteria for at least one disorder. Point prevalence was highest for FM and GAD, followed by IBS, and lowest for MDD and ME/CFS.

3.2. LCA and MM-IRT models

LCA models improved in fit as the number of classes increased (Table 2), primarily reflecting similar symptom patterns differing in severity (Fig. S1). A one-factor EFA supported a general severity dimension underlying ID and FD symptoms (Fig. S2; Table S7). MM-IRT models showed improved fit compared to LCA, and the six-class MM-IRT model was selected based on the lowest BIC, sBIC, and CAIC values and interpretability. The validation set produced comparable fit indices and class proportions and showed adequate classification quality (Table S8–S9). Item parameters and between-class external variable comparisons are reported in the Supplemental Results.

3.3. MM-IRT

In the six-class MM-IRT solution, the largest class was the ‘Healthy’ class, with 57.5% of the participants (Fig. 1; Table 3). Class members showed near zero endorsement probability of criteria-specific symptoms (<0.06), while transdiagnostic symptoms trouble sleeping, fatigue, concentration difficulty, and unrefreshing sleep were endorsed more frequently (0.19–0.28). It had the lowest proportion of women and overall severity profile, including the lowest levels of risk factors,

Table 1
Baseline characteristics for full sample ($n = 72,919$).

Variable	Mean or %	SD or SE	Range	% Miss
<i>Demographics</i>				
Age	49.1	12.4	18.0–91.0	0.0
Sex (female), %	59.2	0.002	–	0.0
<i>Diagnostic status (%)[*]</i>				
MDD	2.8	0.001	–	0.0
GAD	5.7	0.001	–	0.0
ME/CFS	3.0	0.001	–	0.3
FM	6.4	0.001	–	8.7
IBS	5.5	0.001	–	0.2
<i>Risk factors</i>				
BMI	25.9	4.1	12.9–43.3	0.4
Heavy drinking (%)	4.8	0.001	–	0.0
Childhood sexual abuse (%)	7.2	0.001	–	37.5
Chronic stress	1.8	1.8	0–12	7.8
Stressful life events	0.8	1.1	0–12	9.3
General health	2.7	0.8	1–5	7.9
<i>Disability^{**}</i>				
Unable to work	0.9	3.9	0–30	0.0
Bed bound	0.3	1.7	0–30	0.0
Limited work	2.6	6.2	0–30	0.0
Less focused/careful	1.5	4.7	0–30	0.0

Mean provided for continuous variables and percentages (%) for binary variables. SD: standard deviation, for continuous variables. SE: Standard error, for binary variables. % Miss: percentage of missing values for variable. MDD: major depressive disorder; GAD: generalized anxiety disorder; ME/CFS: myalgic encephalomyelitis/chronic fatigue syndrome; FM: fibromyalgia; IBS: irritable bowel syndrome; BMI: body mass index.

^{*} Point prevalence estimates reflect disorder status at a specific assessment time point rather than across the lifespan and are generally lower than lifetime prevalence, particularly for episodic disorders such as MDD and GAD.

^{**} Disability measured in how many days in the past 30 days were participants unable to perform or with limitation due to physical and mental health.

Table 2

LCA and MM-IRT model comparisons.

Model	N par	LL	BIC	AIC	CAIC	sBIC	Entropy	Smallest class
<i>LCA (n = 36,459)</i>								
1-class	27	−313,885.0	628,053.6	627,824.0	628,080.6	627,967.8	1.000	1.00
2-class	55	−269,626.1	539,829.9	539,362.2	539,884.9	539,655.2	0.861	0.28
3-class	83	−261,563.6	523,999.1	523,293.2	524,082.1	523,735.3	0.787	0.12
4-class	111	−255,727.2	512,620.4	511,676.4	512,731.4	512,267.6	0.808	0.09
5-class	139	−253,584.8	508,629.6	507,447.6	508,768.6	508,187.9	0.774	0.07
6-class	167	−251,898.1	505,550.4	504,130.2	505,717.4	505,019.7	0.771	0.04
7-class	195	−250,854.2	503,756.7	502,098.4	503,951.7	503,137.0	0.762	0.03
8-class	223	−249,959.3	502,261.0	500,364.6	502,484.0	501,552.3	0.755	0.03
9-class	251	−249,356.4	501,349.2	499,214.7	501,600.2	500,551.5	0.752	0.02
10-class	279	−248,972.1	500,874.8	498,502.2	501,153.8	499,988.1	0.749	0.02
11-class	307	−248,628.2	500,481.1	497,870.4	500,788.1	499,505.5	0.733	0.02
12-class	335	−248,341.1	500,200.9	497,352.1	500,535.9	499,136.3	0.731	0.01
13-class	363	−248,131.7	500,076.4	496,989.4	500,439.4	498,922.8	0.707	0.01
<i>MM-IRT (n = 36,459)</i>								
1-class	54	−261,075.1	522,717.4	522,258.2	522,771.4	522,545.8	1.000	1.00
2-class	109	−251,916.2	504,977.3	504,050.4	505,086.3	504,630.9	0.716	0.19
3-class	164	−249,576.7	500,876.0	499,481.3	501,040.0	500,354.8	0.743	0.17
4-class	219	−248,776.9	499,854.2	497,991.9	500,073.2	499,158.2	0.660	0.14
5-class	274	−248,114.3	499,106.8	496,776.7	499,380.8	498,236.0	0.690	0.06
6-class	329	−247,630.9	498,717.6	495,919.8	499,046.6	497,672.0	0.678	0.04
7-class	384	−247,481.8	498,997.2	495,731.7	499,381.2	497,776.8	0.635	0.05
8-class	439	−247,237.6	499,086.4	495,353.2	499,525.4	497,691.3	0.626	0.03
9-class	494	−247,026.6	499,242.1	495,041.1	499,736.1	497,672.2	0.603	0.03
<i>MM-IRT validation (n = 36,460)</i>								
6-class		−247,719.6	498,895.0	496,097.2	499,224.0	497,849.4	0.674	0.05

The LCA and MM-IRT were estimated in the training set ($n = 36,459$) and the final MM-IRT model was validated in the test set ($n = 36,460$). N par: number of parameters; LL: Log Likelihood; BIC: Bayesian information criterion; AIC: Akaike information criterion; CAIC: Consistent AIC; sBIC: sample size adjusted BIC. Entropy is a measure of separation between latent classes, where higher values indicate better class separation.

unemployment, and disability days. Participants in this class did not meet diagnostic criteria for IDs or FDs, except for 2.0% reporting IBS.

Class 2 ('Pain', 13.8%) was mainly defined by ME/CFS and FM pain symptoms, with near zero endorsement of ID symptoms (<0.08). The highest endorsed symptoms were joint pain (0.76), musculoskeletal pain (0.49), and muscle pain (0.48). This class also showed the highest endorsement of the ME/CFS symptom PEM (0.39), relative to the other classes. Participants were comparatively older, had higher BMI, and met criteria only for FDs, most commonly FM (18.5%).

Class 3 ('Tension/Pain', 9.1%) showed a similar pain symptom pattern to the Pain class, but at lower endorsement levels. This class was characterized by high endorsement of GAD symptoms muscle tension, restlessness, and irritability (0.41–0.87), as well as transdiagnostic symptoms (0.41–0.56). This class had the highest proportion of women (70.8%). FD prevalence was similar to the Pain class, with 5.0% also meeting GAD criteria.

Class 4 ('Anxiety', 9.0%) was characterized by high endorsement of all seven GAD criteria symptoms. The most frequently reported were muscle tension (0.90), fatigue (0.82), and restlessness (0.70). Participants were comparatively younger (mean 42.4 years), more likely employed (86.9%), and reported high chronic stress. It had the highest prevalence of GAD (39.5%) and greatest proportion of MDD-GAD only comorbidity (7.9%). Compared to the Tension/Pain class, it showed higher MDD specific symptom endorsement and lower FD specific symptom endorsement. Correspondingly, 15.4% met criteria for MDD, whereas none met criteria for ME/CFS or FM.

Class 5 ('Cognition/Fatigue', 6.4%) was defined by high endorsement of concentration difficulty (1.00), fatigue (0.84), difficulty thinking (0.80), and unrefreshing sleep (0.73). Moderate GAD (<0.58) and pain (<0.31) symptoms and minimal MDD symptoms (<0.15) were endorsed. This class showed the highest prevalence of FDs compared with the other classes.

Class 6 ('Depression', 4.3%) showed the highest endorsement of MDD symptoms across classes, particularly anhedonia (0.51), depressed mood (0.39), and appetite change (0.24). Moderate GAD symptoms (restlessness, muscle tension, and irritability, 0.30–0.44) and moderate-

to-high transdiagnostic symptoms (0.45–0.67) were also reported. FD symptoms were modestly endorsed (<0.22). MDD prevalence (29.8%), disability-related unemployment (11.0%), and disability days were highest in this class, with greatest limitations in work/activities or productivity. The Cognition/Fatigue and Depression classes had approximately 16.0% of participants meeting criteria for at least one ID and one FD, the highest across all classes.

Transdiagnostic symptoms were endorsed across all classes, with the highest prevalence in the Cognition/Fatigue class. IBS symptom endorsement was similar across classes (0.12–0.16), except in the Healthy class, where it was lowest. ME/CFS symptoms sore throat and tender lymph nodes showed near zero endorsement across all classes (<0.03). Item parameters reflected these patterns, with transdiagnostic symptoms showing low difficulty and discrimination, and sore throat, tender lymph nodes, and IBS symptom showing high difficulty and low discrimination (Tables S10–S11).

4. Discussion

Hybrid mixture modeling was used to identify six classes of individuals based on ID and FD symptoms while capturing individual-level variation in symptom severity in a large general population sample. Resulting classes reflected distinct but partially overlapping symptom profiles, including predominantly FD-related (Pain), ID-related (Anxiety), and mixed presentations (Tension/Pain, Cognition/Fatigue, Depression), alongside a large low-symptom (Healthy) group. Transdiagnostic symptoms, including fatigue, sleep problems, and concentration difficulty, were consistently endorsed across all classes. ID-FD comorbidity was highest in the Cognition/Fatigue and Depression classes, with the latter showing greatest functional impairment. Our findings support a hybrid conceptualization requiring both categorical and dimensional features to capture heterogeneity in ID and FD symptoms.

To our knowledge, no prior study has investigated ID and FD symptoms using a hybrid mixture modeling approach. However, our results align with latent class studies finding classes characterized by a

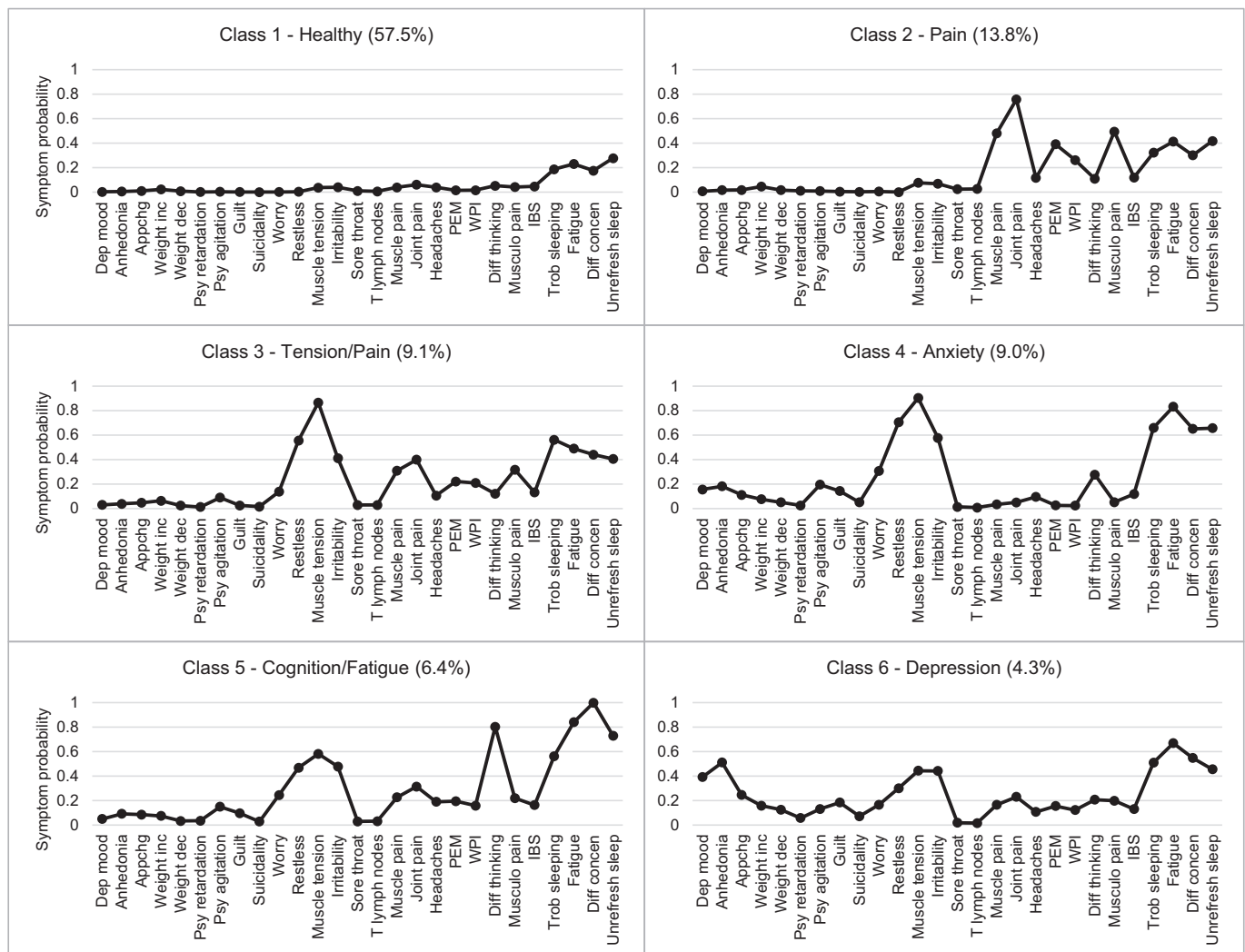


Fig. 1. Symptom profiles in the final 6-class MM-IRT model. Dep mood: depressed mood; Appchg: appetite change; weight inc: weight increase; weight dec: weight decrease; Psy: psychomotor; T lymph nodes: tender lymph nodes; PEM: post-exertional malaise; WPI: widespread pain index; diff thinking: difficulty thinking; IBS: irritable bowel disease symptom; Trob sleeping: trouble sleeping; Unrefresh sleep: unrefreshing sleep.

combination of symptoms from multiple disorders rather than “pure” classes [16–18,20,23]. Importantly, the classes showed qualitative differences in their symptom patterns and characteristics. The Pain class was defined by FD symptoms, with no ID comorbidity, whereas the Anxiety class was characterized primarily by GAD symptoms, with no FD comorbidity. Age patterns were consistent with prior findings, with subgroups defined by high FD pain symptoms tending to be older [19] and those defined by GAD symptoms tending to be younger [18,23]. In contrast, the Cognition/Fatigue and Depression classes were mixed symptom profiles with high ID-FD comorbidity. The Depression class reporting highest disability is consistent with prior evidence showing greater disability days in individuals with depression compared to those with anxiety [75]. Furthermore, no cases of FD-only comorbidity were observed in this class, suggesting FD comorbidity was always accompanied by MDD. A similar pattern was observed in another study, which reported a higher prevalence of depression and functional impairment in individuals with comorbid FDs compared to those with a single FD [76].

Although the IBS symptom was not highly endorsed in any class, IBS cases were present across all classes, indicating that IBS was not strongly linked to any specific class. This aligns with prior work showing that IBS formed a separate symptom cluster in a network model [11] and a weakly correlated latent dimension in a factor model [10]. Other studies have also highlighted the relative distinctiveness of IBS through its

weaker associations with other IDs and FDs, larger sex differences, and conceptualization as a disorder of gut–brain interaction [5,77]. These findings suggest that IBS shares relatively few symptoms and clinical characteristics with the other disorders examined.

4.1. Non-specific symptoms and comorbidity

ID-FD comorbidity may partly reflect transdiagnostic or non-specific symptoms lowering the threshold for fulfilling criteria for multiple disorders [78,79]. These symptoms occur across multiple diagnostic categories rather than being disorder-specific. All classes showed relatively high endorsement of the transdiagnostic symptoms. These symptoms are common in the general population and even their moderate endorsement can increase the likelihood of crossing diagnostic thresholds. For example, most individuals in the Cognition/Fatigue class reported at least six out of seven GAD items, three out of nine ME/CFS items, and four out of six FM items. Although the Depression class showed higher overall MDD symptom prevalence, MDD and GAD co-occurred more frequently in the Anxiety class, where GAD and transdiagnostic symptoms were more and MDD symptoms less strongly endorsed. A previous study similarly reported significantly higher transdiagnostic symptom endorsement in participants with MDD-GAD comorbidity than in those with MDD alone [80]. The high endorsement of transdiagnostic

Table 3
6-class MM-IRT characteristics.

Class	Healthy (1)	Pain (2)	Tension/Pain (3)	Anxiety (4)	Cognition/Fatigue (5)	Depression (6)
Size (%)	57.5	13.8	9.1	9.0	6.4	4.3
<i>Demographics</i>						
Age	49.1	55.1	50.4	42.4	47.2	46.3
Sex (female), %	53.8	67.6	70.8	66.7	70.2	58.0
<i>Diagnostic status (%)</i>						
MDD	0.0	0.0	0.0	15.4	3.6	29.8
GAD	0.0	0.0	5.0	39.5	24.6	9.8
ME/CFS	0.0	7.6	10.2	0.0	11.4	8.8
FM	0.0	18.5	20.3	0.0	22.4	15.1
IBS	2.0	10.3	11.0	6.4	13.2	11.4
MDD + GAD only	0.0	0.0	0.0	7.9	0.0	1.9
≥1 ID + ≥1 FD	0.0	0.0	5.9	2.6	16.9	16.3
MDD + any FD	0.0	0.0	0.0	0.2	0.5	7.7
GAD + any FD	0.0	0.0	6.3	0.0	11.4	0.0
≥2 FDs*	0.0	7.4	6.6	0.0	2.2	0.0
<i>Risk factors</i>						
BMI	25.6	27.3	26.1	25.2	26.1	27.0
Heavy drinking (%)	4.5	4.3	4.7	6.0	5.0	6.4
Childhood sexual abuse (%)	4.0	9.7	13.2	8.5	15.0	14.3
Chronic stress	1.2	1.7	2.5	3.4	3.3	2.9
Stressful life events	0.6	0.9	1.0	1.1	1.1	1.3
General health	2.3	3.1	3.0	2.8	3.3	3.3
<i>Employment status (%)</i>						
Employed	79.7	59.9	70.3	86.9	73.7	64.2
Unemployed	2.7	5.4	5.0	6.3	4.7	7.5
Unemployed (disability)	0.4	8.8	6.9	2.1	9.1	11.0
<i>Disability**</i>						
No daily activities	0.1	1.9	1.4	1.7	2.8	4.2
Remain in bed	0.1	0.5	0.5	0.6	0.9	1.5
Limited activities	0.4	4.8	4.3	5.6	7.7	8.8
Less focused/careful	0.1	2.3	2.3	4.0	5.6	5.5

MDD: major depressive disorder; GAD: generalized anxiety disorder; ME/CFS: myalgic encephalomyelitis/chronic fatigue syndrome; FM: fibromyalgia; IBS: irritable bowel syndrome; BMI: body mass index.

MDD + any FD = participants with MDD (and no GAD) with at least one comorbid FD. GAD + any FD = participants with GAD (and no MDD) with at least one comorbid FD.

* Participants with FD comorbidity without MDD or GAD.

** Disability measured in how many days in the past 30 days were participants unable to perform or performed with limitation due to physical and mental health.

symptoms in the Anxiety class, combined with MDD symptoms, anhedonia and psychomotor agitation, may push individuals over diagnostic thresholds even when overall MDD symptom probability is relatively low. Individuals endorsing these symptoms would still meet the required threshold of five out of nine symptoms for an MDD diagnosis. The inflation of observed comorbidity by transdiagnostic symptoms has been demonstrated previously, where MDD-ME/CFS comorbidity decreased from 10.7% to 2.1% when MDD was defined only by its two core symptoms, depressed mood and anhedonia [78].

Etiologically, non-specific symptoms may arise from multiple mechanisms rather than a single pathway, resulting in their high endorsement across disorders. Moreover, their endorsement may reflect different underlying processes depending on the clinical context and can complicate diagnosis. For example, somatic MDD symptoms, such as fatigue and sleep disturbances, may reflect pain rather than emotional distress in individuals with FM, allowing them to meet MDD thresholds despite endorsing few core depressive features [81]. Similarly, ME/CFS pain symptoms frequently co-occur in FM, potentially contributing to higher FM prevalence across classes, whereas sore throat and tender lymph nodes are rarely endorsed and are weak indicators of ME/CFS [10,76]. These findings indicate that some disorder-defining symptoms (e.g., fatigue and headaches) capture general malaise rather than disorder-specific pathology [10].

4.2. Strengths and limitations

A major strength of this study is the large general population sample size, allowing estimation and validation of the MM-IRT model in independent subsamples. Additionally, ID and FD symptoms were assessed

according to diagnostic criteria within the same wave rather than across the lifetime, enabling identification of transdiagnostic subtypes while minimizing recall bias. This study also has limitations. First, symptoms were dichotomized, potentially resulting in some loss of information from the ordinal FD variables. However, the model including a continuous within-class severity factor likely mitigated this limitation by allowing gradation at the person-level based on overall symptom patterns. Second, class assignment was based on highest posterior probability. This can introduce some uncertainty in class membership, as some participants may show clear distinctions while others may show similar probabilities of belonging to multiple classes. Third, IBS was represented by a single binary item, whereas other disorders were modeled using multiple symptoms. This measurement constraint may have contributed to the non-specificity of the IBS symptom. Fourth, we examined symptoms in the context of selected IDs and FDs. In a general population sample, similar symptom profiles may also arise from unmeasured conditions (e.g., trauma-related disorders or medical conditions). However, a previous general population study found that FD symptom classes remained largely unchanged after excluding individuals with chronic diseases [17], suggesting that such influences are unlikely to substantially alter overall class patterns. Lastly, participants in Lifelines are predominantly native Dutch and live in the north of the Netherlands, potentially limiting generalizability of our findings to populations with different cultural and ethnic backgrounds, where ID and FD symptom patterns may differ [82,83].

4.3. Implications and future research

This study illustrates how small differences in co-occurring symptom

patterns can yield substantial differences in diagnostic prevalence. Although the Tension/Pain and Anxiety classes showed similar GAD symptom endorsement, GAD prevalence was much higher in the Anxiety class (5.0% vs 39.5%). This likely reflects low endorsement of excessive worry in the Tension/Pain class, which is required for diagnosis. It is possible that some individuals in this class may have subthreshold GAD despite high average endorsement. In the Anxiety class, GAD symptoms co-occurred with additional ID and transdiagnostic symptoms, increasing the likelihood of meeting full criteria, whereas in the Tension/Pain class they appeared alongside FD symptoms and lower transdiagnostic symptom endorsement, reducing the probability of crossing the diagnostic threshold. This categorical approach overlooks the continuous nature of symptoms and clinically meaningful sub-threshold features that may still be associated with substantial disability [81,84,85].

MM-IRT results showed that symptoms vary in their contribution to subgroup differentiation, with differences in severity thresholds and discrimination. Transdiagnostic symptoms and ME/CFS symptoms sore throat and tender lymph nodes provided limited information for identifying meaningful profiles; transdiagnostic symptoms were common across severity levels and sore throat and tender lymph nodes rare and only emerging at higher severity. These findings have implications for diagnostic frameworks, particularly ME/CFS, for which multiple case definitions use different symptom sets. Using different case definitions may influence the proportion of individuals meeting ME/CFS criteria across classes [86]. Therefore, evaluating individual symptom contributions is important for refining classifications and identifying clinically meaningful subgroups.

With these six data-driven symptom-based classes, we do not yet propose replacing current diagnostic criteria for IDs and FDs. For accurate classification of these disorders, it is necessary to consider validators alongside symptom-based criteria, such as familial incidence, biological markers, and treatment response [87]. Future studies incorporating additional validators are needed to clarify mechanisms underlying comorbidity and to evaluate the clinical utility of these classes, including their ability to predict prognosis and functional outcomes.

Further research is necessary to understand ID-FD comorbidity. Transdiagnostic symptoms were consistently endorsed across classes and may act as bridge symptoms connecting IDs and FDs, potentially triggering or exacerbating other symptoms [11,88]. Longitudinal studies are needed to examine how shared vulnerabilities develop and contribute to comorbidity. Additionally, the Cognition/Fatigue and Depression classes showed high FD comorbidity with GAD and MDD, respectively. This could suggest potential bidirectional relationships between IDs and FDs [89,90], although this cannot be determined due to the cross-sectional design. Future longitudinal studies should assess class stability and identify predictors of ID-FD comorbidity. Methodologically, we estimated unidimensional MM-IRT models and, based on prior evidence for the multidimensional structure of ID and FD symptoms [10], future work could extend this framework by including multiple within-class latent traits to examine whether they capture meaningful constructs.

5. Conclusion

Using MM-IRT, we identified six ID and FD symptom classes in a large general population sample: Healthy, Pain, Tension/Pain, Anxiety, Cognition/Fatigue, and Depression classes. Non-specific symptoms may contribute to ID-FD comorbidity by lowering the threshold for meeting criteria of multiple disorders. Empirically derived symptom profiles can inform diagnostic refinement by revealing differences in the clinical utility of individual symptoms, while longitudinal research is needed to clarify the mechanisms underlying transdiagnostic symptoms.

Ethical standard

The Lifelines cohort study followed the guidelines of the Declaration of Helsinki, and has been approved by the UMCG Medical ethical committee under number 2007/152. All participants provided written informed consent for participation.

CRediT authorship contribution statement

Urvi Saini: Writing – original draft, Visualization, Methodology, Formal analysis, Data curation, Conceptualization. **Rob Wanders:** Writing – review & editing, Methodology. **Albertine J. Oldehinkel:** Writing – review & editing, Supervision, Conceptualization. **Judith G. M. Rosmalen:** Writing – review & editing, Supervision, Conceptualization. **Hanna M. van Loo:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Data curation, Conceptualization.

Funding statement

This project was supported by NIMH grant R01MH125902. The Lifelines initiative has been made possible by subsidy from the Dutch Ministry of Health, Welfare and Sport, the Dutch Ministry of Economic Affairs, the University Medical Center Groningen, Groningen University, and the Provinces in the North of the Netherlands (Drenthe, Friesland, Groningen).

Declaration of competing interest

The authors declare that they have no competing interests.

Acknowledgments

We thank the members of the Pathways of Functional and Internalizing Disorders (PATHFINDER) consortium for their contributions to this project. The authors also wish to acknowledge the services of the Lifelines Cohort Study, the contributing research centers delivering data to Lifelines, and all the study participants. No generative artificial intelligence tools were used in any aspect of the literature review, data collection, data analysis, drafting, or revising of the manuscript.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.genhosppsy.2026.08.006>.

Data availability

Researchers can apply to use the Lifelines data used in this study. Information about how to request data and the conditions of use can be found on their website (<https://www.lifelines-biobank.com>).

References

- [1] Frye SS, Perfect MM, Graham JW. Internalizing Disorders. In: Forman S, Shahidullah J, editors. Handbook of pediatric behavioral healthcare. Cham: Springer; 2018. https://doi.org/10.1007/978-3-030-00791-1_12.
- [2] Belzer K, Schneier FR. Comorbidity of anxiety and depressive disorders: issues in conceptualization, assessment, and treatment. *J Psychiatr Pract* 2004;10(5): 296–306. <https://doi.org/10.1097/00131746-200409000-00003>.
- [3] Ter Meulen WG, Draisma S, van Hemert AM, Schoevers RA, Kupka RW, Beekman ATF, et al. Depressive and anxiety disorders in concert-A synthesis of findings on comorbidity in the NESDA study. *J Affect Disord* 2021;284:85–97. <https://doi.org/10.1016/j.jad.2021.02.004>.
- [4] Buffington CA. Developmental influences on medically unexplained symptoms. *Psychother Psychosom* 2009;78(3):139–44. <https://doi.org/10.1159/000206866>.
- [5] Thomas NS, Gillespie NA, Kendler KS, Oldehinkel AJ, Rosmalen JGM, van Loo HM. Comorbidity and sex differences in functional disorders and internalizing disorders. *Gen Hosp Psychiatry* 2024;90:91–8. <https://doi.org/10.1016/j.genhosppsy.2024.07.013>.

- [6] Baumeister H, Balke K, Härter M. Psychiatric and somatic comorbidities are negatively associated with quality of life in physically ill patients. *J Clin Epidemiol* 2005;58(11):1090–100. <https://doi.org/10.1016/j.jclinepi.2005.03.011>.
- [7] Bos M, Monden R, Wray NR, Zhou Y, Kendler KS, Rosmalen JGM, et al. Familial coaggregation and shared familiarity of functional and internalizing disorders in the lifelines cohort. *Psychol Med* 2025;55:e126. <https://doi.org/10.1017/S003329172500100X>.
- [8] Chaves-Filho AJM, Macedo DS, de Lucena DF, Maes M. Shared microglial mechanisms underpinning depression and chronic fatigue syndrome and their comorbidities. *Behav Brain Res* 2019;372:111975. <https://doi.org/10.1016/j.bbr.2019.111975>.
- [9] Kato K, Sullivan PF, Evengård B, Pedersen NL. A population-based twin study of functional somatic syndromes. *Psychol Med* 2009;39(3):497–505. <https://doi.org/10.1017/S0033291708003784>.
- [10] Saini U, Aggen SH, Oldehinkel AJ, Rosmalen JGM, van Loo HM. Disentangling comorbidity: symptom dimensions of internalizing and functional disorders in a large general population sample. *BMC Psychiatry* 2026. <https://doi.org/10.1186/s12888-026-07797-5>. Advance online publication. doi:10.1186/s12888-026-07797-5.
- [11] Saini U, Rosmalen JGM, Oldehinkel AJ, van Loo HM. Connecting the dots: network structures of internalizing and functional symptoms in a population-based cohort. *J Psychosom Res* 2024;187:111932. <https://doi.org/10.1016/j.jpsychores.2024.111932>.
- [12] Vassend O, Røysamb E, Nielsen CS, Czajkowski NO. Fatigue symptoms in relation to neuroticism, anxiety-depression, and musculoskeletal pain. A longitudinal twin study. *PLoS One* 2018;13(6):e0198594. <https://doi.org/10.1371/journal.pone.0198594>.
- [13] Martínez MP, Sánchez AI, Prados G, Lami MJ, Villar B, Miró E. Fibromyalgia as a heterogeneous condition: subgroups of patients based on physical symptoms and cognitive-affective variables related to pain. *Span J Psychol* 2021;24:e33. <https://doi.org/10.1017/SJP.2021.30>.
- [14] Nandi A, Beard JR, Galea S. Epidemiologic heterogeneity of common mood and anxiety disorders over the lifespan in the general population: a systematic review. *BMC Psychiatry* 2009;9:31. <https://doi.org/10.1186/1471-244X-9-31>.
- [15] Williams TE, Chalder T, Sharpe M, White PD. Heterogeneity in chronic fatigue syndrome - empirically defined subgroups from the PACE trial. *Psychol Med* 2017;47(8):1454–65. <https://doi.org/10.1017/S0033291716003615>.
- [16] Beijers L, Wardenaar KJ, Bosker FJ, Lamers F, van Groothedest G, de Boer MK, et al. Biomarker-based subtyping of depression and anxiety disorders using latent class analysis. A NESDA study. *Psychol Med* 2019;49(4):617–27. <https://doi.org/10.1017/S0033291718001307>.
- [17] Eliassen M, Schröder A, Fink P, Kreiner S, Dantoft TM, Poulsen CH, et al. A step towards a new delimitation of functional somatic syndromes: A latent class analysis of symptoms in a population-based cohort study. *J Psychosom Res* 2018;108:102–17. <https://doi.org/10.1016/j.jpsychores.2018.03.002>.
- [18] Lei H, Chen C, Hagiwara K, Kusumi I, Tanabe H, Inoue T, et al. Symptom patterns of the occurrence of depression and anxiety in a Japanese general adult population sample: A latent class analysis. *Front Psych* 2022;13:808918. <https://doi.org/10.3389/fpsy.2022.808918>.
- [19] Schmalbach B, Häuser W, Brähler E, Henningsen P, & Wolfe F. (2021). Fibromyalgia revisited: do latent class analyses of symptom profiles in the general population confirm 2016 fibromyalgia diagnostic criteria? *Clin Exp Rheumatol*, 39 Suppl 130(3), 128–136. doi:10.55563/clinexpheumatol/bexjrm.
- [20] Sullivan PF, Smith W, Buchwald D. Latent class analysis of symptoms associated with chronic fatigue syndrome and fibromyalgia. *Psychol Med* 2002;32(5):881–8. <https://doi.org/10.1017/s0033291702005834>.
- [21] van Loo HM, Wanders RBK, Wardenaar KJ, Fried EI. Problems with latent class analysis to detect data-driven subtypes of depression. *Mol Psychiatry* 2018;23(3):495–6. <https://doi.org/10.1038/mp.2016.202>.
- [22] Sunderland M, Carragher N, Wong N, Andrews G. Factor mixture analysis of DSM-IV symptoms of major depression in a treatment seeking clinical population. *Compr Psychiatry* 2013;54(5):474–83. <https://doi.org/10.1016/j.comppsych.2012.12.011>.
- [23] Wanders RB, van Loo HM, Vermunt JK, Meijer RR, Hartman CA, Schoevers RA, et al. Casting wider nets for anxiety and depression: disability-driven cross-diagnostic subtypes in a large cohort. *Psychol Med* 2016;46(16):3371–82. <https://doi.org/10.1017/S0033291716002221>.
- [24] Wardenaar KJ, Wanders RBK, Ten Have M, de Graaf R, de Jonge P. Using a hybrid subtyping model to capture patterns and dimensionality of depressive and anxiety symptomatology in the general population. *J Affect Disord* 2017;215:125–34. <https://doi.org/10.1016/j.jad.2017.03.038>.
- [25] Blanco C, Rubio J, Wall M, Wang S, Jiu CJ, Kendler KS. Risk factors for anxiety disorders: common and specific effects in a national sample. *Depress Anxiety* 2014;31(9):756–64. <https://doi.org/10.1002/da.22247>.
- [26] Dagnino P, Ugarte MJ, Morales F, González S, Saralegui D, Ehrenthal JC. Risk factors for adult depression: adverse childhood experiences and personality functioning. *Front Psychol* 2020;11:594698. <https://doi.org/10.3389/fpsyg.2020.594698>.
- [27] Monden R, Rosmalen JGM, Wardenaar KJ, Creed F. Predictors of new onsets of irritable bowel syndrome, chronic fatigue syndrome and fibromyalgia: the lifelines study. *Psychol Med* 2022;52(1):112–20. <https://doi.org/10.1017/S0033291720001774>.
- [28] Scholtens S, Smidt N, Swertz MA, Bakker SJ, Dotinga A, Vonk JM, et al. Cohort profile: Lifelines, a three-generation cohort study and biobank. *Int J Epidemiol* 2015;44(4):1172–80. <https://doi.org/10.1093/ije/dy229>.
- [29] Klijs B, Scholtens S, Mandemakers JJ, Snieder H, Stolk RP, Smidt N. Representativeness of the Lifelines cohort study. *PLoS One* 2015;10(9):e0137203. <https://doi.org/10.1371/journal.pone.0137203>.
- [30] Sijtsma A, Rienks J, van der Harst P, Navis G, Rosmalen JGM, Dotinga A. Cohort profile update: lifelines, a three-generation cohort study and biobank. *Int J Epidemiol* 2022;51(5):e295–302. <https://doi.org/10.1093/ije/dyab257>.
- [31] Sheehan DV, Lecrubier Y, Sheehan KH, Amorim P, Janav S, Weiller E, et al. The Mini-international neuropsychiatric interview (M.I.N.I.): the development and validation of a structured diagnostic psychiatric interview for DSM-IV and ICD-10. *J Clin Psychiatry* 1998;59:22–33.
- [32] Fukuda K, Straus SE, Hickie I, Sharpe MC, Dobbins JG, Komaroff A, et al. The chronic fatigue syndrome: a comprehensive approach to its definition and study. *Ann Intern Med* 1994;121:953–9.
- [33] Wolfe F, Clauw DJ, Fitzcharles MA, Goldenberg DL, Katz RS, Mease P, et al. The American College of Rheumatology preliminary diagnostic criteria for fibromyalgia and measurement of symptom severity. *Arthritis Care Res* 2010;62(5):600–10. <https://doi.org/10.1002/acr.20140>.
- [34] Vercoulen JH, Swanink CM, Fennis JF, Galama JM, van der Meer JW, Bleijenberg G. Dimensional assessment of chronic fatigue syndrome. *J Psychosom Res* 1994;38(5):383–92. [https://doi.org/10.1016/0022-3999\(94\)90099-x](https://doi.org/10.1016/0022-3999(94)90099-x).
- [35] Drossman DA. The functional gastrointestinal disorders and the Rome III process. *Gastroenterology* 2006;130(5):1377–90.
- [36] Drossman DA. Functional gastrointestinal disorders: what's new for Rome IV? *Lancet Gastroenterol Hepatol* 2016;1(1):6–8. [https://doi.org/10.1016/S2468-1253\(16\)30022-X](https://doi.org/10.1016/S2468-1253(16)30022-X).
- [37] Verhulst B, Neale MC. Best practices for binary and ordinal data analyses. *Behav Genet* 2021;51(3):204–14. <https://doi.org/10.1007/s10519-020-10031-x>.
- [38] Bakken IJ, Tveito K, Gunnes N, Ghaderi S, Stoltenberg C, Trostad L, et al. Two age peaks in the incidence of chronic fatigue syndrome/myalgic encephalomyelitis: a population-based registry study from Norway 2008–2012. *BMC Med* 2014;12:167. <https://doi.org/10.1186/s12916-014-0167-5>.
- [39] Creed F. Review article: the incidence and risk factors for irritable bowel syndrome in population-based studies. *Aliment Pharmacol Ther* 2019;50(5):507–16. <https://doi.org/10.1111/apt.15396>.
- [40] Creed F. A review of the incidence and risk factors for fibromyalgia and chronic widespread pain in population-based studies. *Pain* 2020;161(6):1169–76. <https://doi.org/10.1097/j.pain.0000000000001819>.
- [41] van Loo HM, Beijers L, Wieling M, de Jong TR, Schoevers RA, Kendler KS. Prevalence of internalizing disorders, symptoms, and traits across age using advanced nonlinear models. *Psychol Med* 2023;53(1):78–87. <https://doi.org/10.1017/S0033291721001148>.
- [42] Olesen SC, Butterworth P, Leach LS, Kelaher M, Pirkis J. Mental health affects future employment as job loss affects mental health: findings from a longitudinal population study. *BMC Psychiatry* 2013;13:144. <https://doi.org/10.1186/1471-244X-13-144>.
- [43] Schovsbo SU, Dantoft TM, Thuesen BH, Leth-Møller KB, Eplow LF, Petersen MW, et al. Social position and functional somatic disorders: the DanFunD study. *Scand J Public Health* 2023;51(2):225–32. <https://doi.org/10.1177/14034948211056752>.
- [44] Zhao G, Ford ES, Dhingra S, Li C, Strine TW, Mokdad AH. Depression and anxiety among US adults: associations with body mass index. *International journal of obesity* (2005) 2009;33(2):257–66. <https://doi.org/10.1038/ijo.2008.268>.
- [45] Hsu TY, He GY, Wang YC, Chen CY, Wang SH, Chen WK, et al. Alcohol use disorder increases the risk of irritable bowel disease: A Nationwide retrospective cohort study. *Medicine* 2015;94(51):e2334. <https://doi.org/10.1097/MD.0000000000002334>.
- [46] Lai HM, Cleary M, Sitharthan T, Hunt GE. Prevalence of comorbid substance use, anxiety and mood disorders in epidemiological surveys, 1990–2014: A systematic review and meta-analysis. *Drug Alcohol Depend* 2015;154:1–13. <https://doi.org/10.1016/j.drugalcdep.2015.05.031>.
- [47] Malykhin N, Serrano J, Pietrasik W, Hegadoren K. Effects of duration and intensity of psychological stressors on mental health outcomes. *J Psychiatr Res* 2025;187:211–22. <https://doi.org/10.1016/j.jpsychores.2025.05.012>.
- [48] Bjørndal LD, Ebrahimi OV, Røysamb E, Karstoft KI, Czajkowski NO, Nes RB. Stressful life events exhibit complex patterns of associations with depressive symptoms in two population-based samples using network analysis. *J Affect Disord* 2024;349:569–76. <https://doi.org/10.1016/j.jad.2024.01.054>.
- [49] Miloyan B, Joseph Bienvenu O, Brilot B, Eaton WW. Adverse life events and the onset of anxiety disorders. *Psychiatry Res* 2018;259:488–92. <https://doi.org/10.1016/j.psychres.2017.11.027>.
- [50] De Venter M, Illegems J, Van Royen R, Moorkens G, Sabbe BGC, Van Den Eede F. Differential effects of childhood trauma subtypes on fatigue and physical functioning in chronic fatigue syndrome. *Compr Psychiatry* 2017;78:76–82. <https://doi.org/10.1016/j.comppsych.2017.07.006>.
- [51] Kaleycheva N, Cullen AE, Evans R, Harris T, Nicholson T, Chalder T. The role of lifetime stressors in adult fibromyalgia: systematic review and meta-analysis of case-control studies. *Psychol Med* 2021;51(2):177–93. <https://doi.org/10.1017/S0033291720004547>.
- [52] Lee AH, Mahurkar-Joshi S, Naliboff B, Gupta A, Labus J, Tillisch K, et al. Role of sex, anxiety, and resilience in the association between adverse childhood experiences and irritable bowel syndrome. *Clin. Gastroenterol. Hepatol.* 2025;23(1):154–162.e2. <https://doi.org/10.1016/j.cgh.2024.05.041>.
- [53] Östberg D, Nordin S. Three-year prediction of depression and anxiety with a single self-rated health item. *J Ment Health* 2022;31(3):402–9. <https://doi.org/10.1080/09638237.2021.2022610>.
- [54] Sattel H, Häuser W, Schmalbach B, Brähler E, Henningsen P, Hausteiner-Wiehle C. Functional somatic disorders, their subtypes, and their association with self-rated

- health in the German general population. *Psychosom Med* 2023;85(4):366–75. <https://doi.org/10.1097/PSY.0000000000001187>.
- [55] Comer JS, Blanco C, Hasin DS, Liu SM, Grant BF, Turner JB, et al. Health-related quality of life across the anxiety disorders: results from the national epidemiologic survey on alcohol and related conditions (NESARC). *J Clin Psychiatry* 2011;72(1): 43–50. <https://doi.org/10.4088/JCP.09m05094blu>.
- [56] Joustra ML, Janssens KA, Bültmann U, Rosmalen JG. Functional limitations in functional somatic syndromes and well-defined medical diseases. Results from the general population cohort LifeLines. *J Psychosom Res* 2015;79(2):94–9. <https://doi.org/10.1016/j.jpsychores.2015.05.004>.
- [57] Shumye S, Belayneh Z, Mengistu N. Health related quality of life and its correlates among people with depression attending outpatient department in Ethiopia: a cross sectional study. *Health Qual Life Outcomes* 2019;17(1):169. <https://doi.org/10.1186/s12955-019-1233-7>.
- [58] Weigel B, Inderyas M, Eaton-Fitch N, Thapaliya K, Marshall-Gradisnik S. Health-related quality of life in Myalgic encephalomyelitis/chronic fatigue syndrome and post COVID-19 condition: a systematic review. *J Transl Med* 2025;23(1):318. <https://doi.org/10.1186/s12967-025-06131-z>.
- [59] Visser M, Depaoli S. A guide to detecting and modeling local dependence in latent class analysis models. *Struct Equ Model Multidiscip J* 2022;29(6):971–82. <https://doi.org/10.1080/10705511.2022.2033622>.
- [60] Enders CK, Bandalos DL. The relative performance of full information maximum likelihood estimation for missing data in structural equation models. *Struct. Eq. Model.* 2001;8(3):430–57. https://doi.org/10.1207/S15328007SEM0803_5.
- [61] Weller BE, Bowen NK, Faubert SJ. Latent class analysis: A guide to best practice. *J Black Psychol* 2020;46(4):287–311. <https://doi.org/10.1177/0095798420930932>.
- [62] De Ayala RJ, Santiago SY. An introduction to mixture item response theory models. *J Sch Psychol* 2017;60:25–40. <https://doi.org/10.1016/j.jsp.2016.01.002>.
- [63] Sen S, Cohen AS. Applications of mixture IRT models: A literature review. *Measur.: Interdisciplin. Res. Perspectiv.* 2019;17(4):177–91. <https://doi.org/10.1080/15366367.2019.1583506>.
- [64] Muthén B, Asparouhov T. Item response mixture modeling: application to tobacco dependence criteria. *Addict Behav* 2006;31(6):1050–66. <https://doi.org/10.1016/j.addbeh.2006.03.026>.
- [65] Killian MO, Cimino AN, Weller BE, Hyun Seo C. (2019). A systematic review of latent variable mixture modeling research in social work journals. *J Evid Based Soc Work* 2019;16(2):192–210. <https://doi.org/10.1080/23761407.2019.1577783>.
- [66] Schwarz G. Estimating the dimension of a model. *Annals of Statistics* 1978;6(2). <https://doi.org/10.1214/aos/1176344136>.
- [67] Sclove SL. Application of model-selection criteria to some problems in multivariate analysis. *Psychometrika* 1987;52(3):333–43. <https://doi.org/10.1007/bf02294360>.
- [68] Akaike H. On entropy maximization principle. In: Krishnaiah PR, editor. *Applications of statistics*. North Holland Publishing; 1977. p. 27–41.
- [69] Bozdoğan H. Model selection and Akaike's information criterion (AIC): the general theory and its analytical extensions. *Psychometrika* 1987;52(3):345–70. <https://doi.org/10.1007/bf02294361>.
- [70] Bakk Z, Tekle FB, Vermunt JK. Estimating the association between latent class membership and external variables using Bias-adjusted three-step approaches. *Sociol. Methodol.* 2013;43(1):272–311. <https://doi.org/10.1177/0081175012470644>.
- [71] Bolck A, Croon M, Hagenaars J. Estimating latent structure models with categorical variables: one-step versus three-step estimators. *Political Anal.* 2004;12(1):3–27. <https://doi.org/10.1093/pan/mp001>.
- [72] Vermunt JK. Latent class modeling with covariates: two improved three-step approaches. *Political Anal.* 2017;18(4):450–69. <https://doi.org/10.1093/pan/mpq025>.
- [73] Muthén LK, Muthén BO. *Mplus User's Guide*. 8th ed. Los Angeles, CA: Muthén, & Muthén; 2017.
- [74] Vermunt JK, Magidson J. *Upgrade manual for LatentGOLD basic, advanced, syntax, and choice versions 6.0 and 6.1*. Tilburg. The Netherlands: Statistical Innovations Europe; 2025.
- [75] Löwe B, Spitzer RL, Williams JB, Mussell M, Schellberg D, Kroenke K. Depression, anxiety and somatization in primary care: syndrome overlap and functional impairment. *Gen Hosp Psychiatry* 2008;30(3):191–9. <https://doi.org/10.1016/j.genhosppsych.2008.01.001>.
- [76] Brown MM, Jason LA. Functioning in individuals with chronic fatigue syndrome: increased impairment with co-occurring multiple chemical sensitivity and fibromyalgia. *Dynamic medicine* : DM 2007;6:6. <https://doi.org/10.1186/1476-5918-6-6>.
- [77] Drossman DA, Chang L, Tack J. Disorders of gut-brain interaction and the Rome V process. *Gastroenterology* 2026;170(6):1083–98. <https://doi.org/10.1053/j.gastro.2026.02.014>.
- [78] Janssens KA, Zijlema WL, Joustra ML, Rosmalen JG. Mood and anxiety disorders in chronic fatigue syndrome, fibromyalgia, and irritable bowel syndrome: results from the LifeLines cohort study. *Psychosom Med* 2015;77(4):449–57. <https://doi.org/10.1097/PSY.0000000000000161>.
- [79] van Loo HM, Romeijn JW. Psychiatric comorbidity: fact or artifact? *Theor Med Bioeth* 2015;36(1):41–60. <https://doi.org/10.1007/s11017-015-9321-0>.
- [80] Zbozinek TD, Rose RD, Wolitzky-Taylor KB, Sherbourne C, Sullivan G, Stein MB, et al. Diagnostic overlap of generalized anxiety disorder and major depressive disorder in a primary care sample. *Depress Anxiety* 2012;29(12):1065–71. <https://doi.org/10.1002/da.22026>.
- [81] Veltri A, Scarpellini P, Piccinni A, Conversano C, Giacomelli C, Bombardieri S, et al. Methodological approach to depressive symptoms in fibromyalgia patients. *Clin Exp Rheumatol* 2012;30(6 Suppl 74):136–42.
- [82] Isaac M, Janca A, Burke KC, Costa e Silva JA, Acuda SW, Altamura AC, et al. Medically unexplained somatic symptoms in different cultures. A preliminary report from phase I of the World Health Organization international study of somatoform disorders. *Psychother Psychosom* 1995;64(2):88–93. <https://doi.org/10.1159/000288996>.
- [83] Watkins DC, Assari S, Johnson-Lawrence V. Race and ethnic group differences in comorbid major depressive disorder, generalized anxiety disorder, and chronic medical conditions. *J Racial Ethn Health Disparities* 2015;2(3):385–94. <https://doi.org/10.1007/s40615-015-0085-z>.
- [84] Rodríguez MR, Nuevo R, Chatterji S, Ayuso-Mateos JL. Definitions and factors associated with subthreshold depressive conditions: a systematic review. *BMC Psychiatry* 2012;12:181. <https://doi.org/10.1186/1471-244X-12-181>.
- [85] Zhang R, Peng X, Song X, Long J, Wang C, Zhang C, et al. The prevalence and risk of developing major depression among individuals with subthreshold depression in the general population. *Psychol Med* 2023;53(8):3611–20. <https://doi.org/10.1017/S0033291722000241>.
- [86] Jason LA, Katz BZ, Sunquist M, Torres C, Cotler J, Bhatia S. The prevalence of pediatric Myalgic encephalomyelitis/chronic fatigue syndrome in a community-based sample. *Child & youth care forum* 2020;49(4):563–79. <https://doi.org/10.1007/s10566-019-09543-3>.
- [87] Kendler KS. A history of the DSM-5 scientific review committee. *Psychol Med* 2013; 43(9):1793–800. <https://doi.org/10.1017/S0033291713001578>.
- [88] Borsboom D, Cramer AO. Network analysis: an integrative approach to the structure of psychopathology. *Annu Rev Clin Psychol* 2013;9:91–121. <https://doi.org/10.1146/annurev-clinpsy-050212-185608>.
- [89] Barlow DH, Allen LB, Choate ML. *Toward a unified treatment for emotional disorders*. In: *The neurotic paradox, Volume 1*. Routledge; 2020. p. 141–66.
- [90] Thomas NS, Neale MC, Kendler KS, van Loo HM, Gillespie NA. Prospective associations between major depressive disorder, generalized anxiety disorder, fibromyalgia, and myalgic encephalomyelitis/chronic fatigue syndrome. *Psychol Med* 2025;55:e232. <https://doi.org/10.1017/S0033291725100603>.